

eqannotate

Declarative equation annotations with automatic layout

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Abstract

`eqannotate` is a LaTeX package for labelled display mathematics. Mark a term, declare its meaning, and let the package arrange the annotation. This manual documents the stable v0.1.1 user interface.

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1 Introduction

EqAnnotate is for *display* equations. Its normal workflow is: mark a term, declare a label, then compile to convergence. Automatic layout is the default workflow, with manual placement available when exact control is useful.

2 Installation

Place `eqannotate.sty` beside the main document and write:

```
\usepackage{eqannotate}
```

For a user TeX tree, install it below `tex/latex/eqannotate/`. EqAnnotate runs entirely within LaTeX/TikZ. Requirements are LaTeX2e, `amsmath`, `xcolor`, TikZ with `tikzmark`, `arrows.meta`, and `calc`, plus `expl3/xparse`. The tested engines are pdfLaTeX, LuaLaTeX, and XeLaTeX.

3 Quick start

$$p(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right)$$

|
Scale

|
Mean

Mark identifiers are local to one display. The declarations may follow the formula in any order.

4 Core API

4.1 Marking terms

```
\eqmark[<color>]{<id>}{<math>}
```

The optional color is used by the colorful theme. IDs are semantic names local to one wrapper. Duplicate marks keep the first target and issue a warning.

4.2 Automatic labels

```
\eqannotate[prefer=auto|above|below]{<id>}{<label>}
```

The optional `prefer` value is a soft side preference interpreted by the automatic layout solver. A label with no matching mark is skipped with a warning.

5 Display environments

Four wrappers are available:

annotatedequation A single display.

annotatedalign A multi-row aligned display.

annotatedgather A centered multi-row display.

annotatedmultline A long display split across rows.

They are unnumbered by default. Adding `[numbered]` gives the complete wrapper one outer equation number; ordinary `\label` and `\ref` work as usual.

$$\begin{array}{c}
 \text{Initial residual} \\
 | \\
 r_0 = x_0 - y, \\
 r_{k+1} = r_k - \alpha_k A p_k, \\
 x_{k+1} = x_k + \alpha_k p_k \\
 \begin{array}{cc}
 \text{State update} & \text{Residual correction}
 \end{array}
 \end{array}
 \tag{1}$$

Equation 1 is one numbered aligned block.

$$\begin{array}{c}
 \text{Energy objective} \\
 | \\
 E(x) = D(x) + R(x), \\
 \nabla E(x) = \nabla D(x) + \nabla R(x) \\
 \text{Objective gradient}
 \end{array}$$

$$\begin{array}{c}
 \text{Total training objective} \qquad \text{Adversarial objective} \\
 \begin{array}{c}
 \mathcal{L}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}}[\ell_{\text{rec}}(x; \theta)] + \lambda_1 \mathbb{E}_{z \sim p(z)}[\ell_{\text{adv}}(z; \theta)] \\
 + \lambda_2 \mathcal{L}_{\text{consistency}}(\theta) + \lambda_3 \mathbb{E}_t[\|v_\theta(x_t, t) - u_t\|_2^2] + \lambda_4 \mathcal{R}_{\text{prior}}(\theta) \\
 \text{Prior regularizer}
 \end{array}
 \end{array}$$

The multiline wrapper retains multiline-style first/last-row alignment. Use **annotatedalign** for short expressions with a deliberately aligned break. It also supports the native commands `\shoveleft` and `\shoveright`.

6 Automatic layout

Automatic placement chooses sides, wraps long labels, performs bounded horizontal de-overlap, allocates lanes, routes connectors, observes the active local `\linewidth`, and reserves vertical space.

$$\mathcal{J} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{reg}} + \gamma \mathcal{L}_{\text{aux}} + \rho \mathcal{R}(\theta)$$

Data fidelity Auxiliary objective
Regularization Prior term

7 Themes and callout styles

```
\eqannotatecolortheme{colorful} % or mono
\eqannotatecalloutstyle{leader} % or arrow
```

In `colorful`, marked terms have pastel backgrounds and related labels and connectors share their colors. In `mono`, term backgrounds are removed and callouts are black. A `leader` is a plain connector; an `arrow` points toward the target.

8 Manual placement

```
\eqannotatemanual[<label TikZ options>][<to-path options>]
  {<id>}{<label>}
```

Use manual placement when a formula benefits from exact control:

$$y = f_{\theta}(x) + \lambda g(x)$$

Base model Manually positioned correction

In manual mode, the supplied TikZ options determine the label position and connector route. EqAnnotate continues to provide the active theme, callout style, label masking, and vertical space reservation. If automatic and manual declarations target the same ID, the manual declaration takes precedence and a warning is emitted.

9 Numbering and references

Numbered aligned and gathered wrappers use one equation number for the complete block. `annotatedmultline` follows the same one-number model as `amsmath multline`. Per-row numbering and custom `\tag` are not part of the current interface. Place `\label` inside a numbered wrapper and reference it normally.

10 Compilation and convergence

Use a normal workflow such as:

```
latexmk -pdf main.tex
```

When compiling by hand, rerun while the log reports

```
Rerun LaTeX for optimized annotation spacing.
```

The package uses remembered positions and the `.aux` file, so a fresh document commonly needs multiple passes.

11 Diagnostics

Diagnostics cover a missing mark, duplicate mark, duplicate annotation, automatic/manual duplication, and commands used outside an `EqAnnotate` display. The last case is an explicit package error; the other authoring mistakes have readable warning behavior.

12 Current scope

The current interface supports display equations and excludes inline annotations, per-row numbering for multi-line wrappers, arbitrary `\tag`, and `\intertext`/`\shortintertext` inside its wrappers. Configure non-white page backgrounds explicitly:

```
\renewcommand\eqannotatebackgroundcolor{<color>}
```

13 Complete examples

The distribution includes the following runnable public-interface examples:

- `examples/basic.tex`
- `examples/style-gallery.tex`
- `examples/amsmath-gallery.tex`
- `examples/manual-gallery.tex`

14 License and project links

`EqAnnotate` is available under the MIT License. The [GitHub repository](#) hosts source and releases; report problems through the [issue tracker](#).